IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

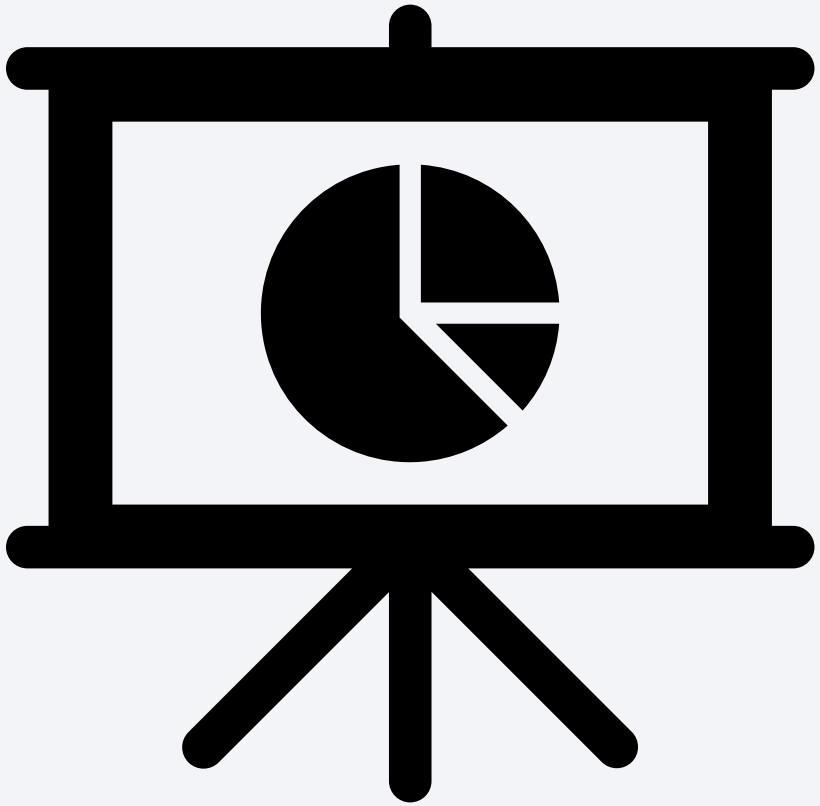
- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Summary of methodologies

The main steps in this project include:

- Data collection
- Data Wrangling
- Exploratory Data Analysis
- Interactive Visual Analytics
- Predictive Analysis



Summary Results:

EDA results with interactive analytics and predictive analysis results.
Github: https://github.com/nikol-damyanova/ibm-capstone-applied-data-science

Introduction

- Project background and context

SpaceX is the leader of commercial space travel. Their Falcon 9 rocket costs about 62 million USD, whereas other providers cost upwards of 165 million USD. The stark difference is due Falcon 9's ability to land, making the first stage of the rocket reusable

- Problems you want to find answers

If we can calculate where the Falcon 9 lands, we can correlate that to its total cost and determine the cost of launch.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:

  Web Scraping

- Perform data wrangling

  Identify and calculate the percentage of the missing values in each attribute

  Use the method value_counts() to determine the number of launches per site, occurrence of each orbit and the outcome

- Perform exploratory data analysis (EDA) using visualization and SQL

  Display the launch sites, payload mass, first successful landing, drone ship landings, successes and failures.

- Perform interactive visual analytics using Folium and Plotly Dash

  Mark launch sites, successes/fails, calculate the distance between point. Present data in an interactive Dash

- Perform predictive analysis using classification models

  How to build, tune, evaluate classification models

Data Collection

For Falcon 9 Launch

Data Collection – SpaceX API

- Request to the SpaceX API
- Clean the requested data

```
spacex_url="https://api.spacexdata.com/v4/launches/past"
```

```
response = requests.get(spacex_url)
```

```
# Lets take a subset of our dataframe keeping only the features we want and the flight number, 
data = data[['rocket', 'payloads', 'launchpad', 'cores', 'flight_number', 'date_utc']]

# We will remove rows with multiple cores because those are falcon rockets with 2 extra rocket 
data = data[data['cores'].map(len)==1]
data = data[data['payloads'].map(len)==1]

# Since payloads and cores are lists of size 1 we will also extract the single value in the lis
data['cores'] = data['cores'].map(lambda x : x[0])
data['payloads'] = data['payloads'].map(lambda x : x[0])

# We also want to convert the date_utc to a datetime datatype and then extracting the date leav
data['date'] = pd.to_datetime(data['date_utc']).dt.date

# Using the date we will restrict the dates of the launches
data = data[data['date'] <= datetime.date(2020, 11, 13)]
```

Data Collection - Scraping

- Web scraping to collect Falcon 9 historical launch records from a Wikipedia page titled `List of Falcon 9 and Falcon Heavy launche`

```
static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_laun

headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
                  "AppleWebKit/537.36 (KHTML, like Gecko) "
                  "Chrome/91.0.4472.124 Safari/537.36"
}
```

```
data=requests.get(static_url).text
```

```
soup=BeautifulSoup(data, 'html.parser')
```

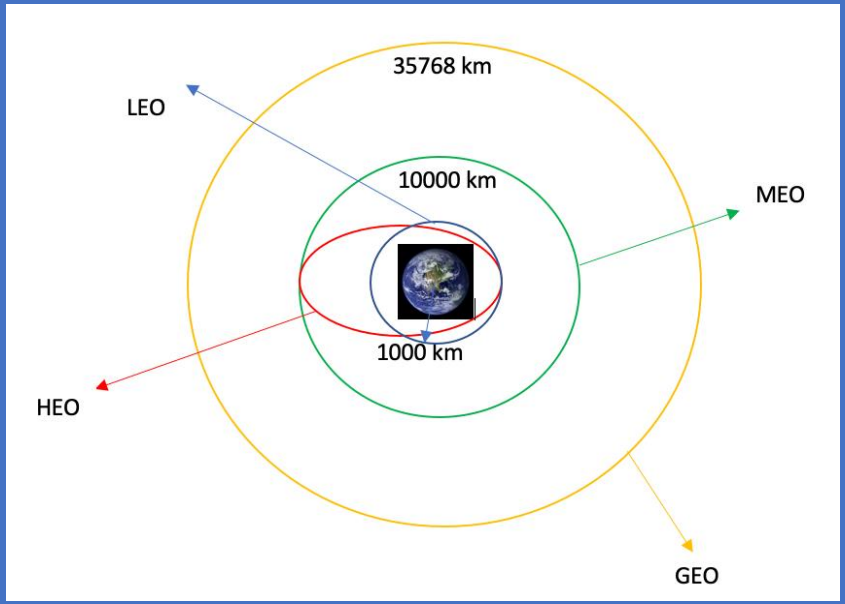
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose

```
launch_dict= dict.fromkeys(column_names)

# Remove an irrelvant column
del launch_dict['Date and time ( )']
```

Data Wrangling

SpaceX has several launch sites stored in the dataset. Using the `.value_counts()` method, we can determine the number of launches, occurances in each orbit and more.



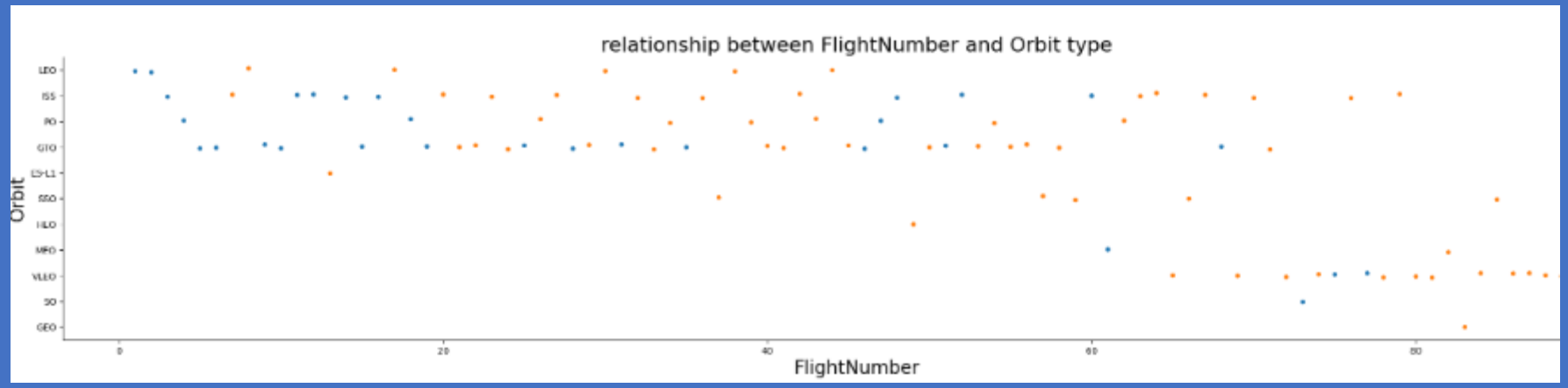
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean

EDA with Data Visualization

SCATTER CHARTS

Scatter charts helped visualize the correlation between:

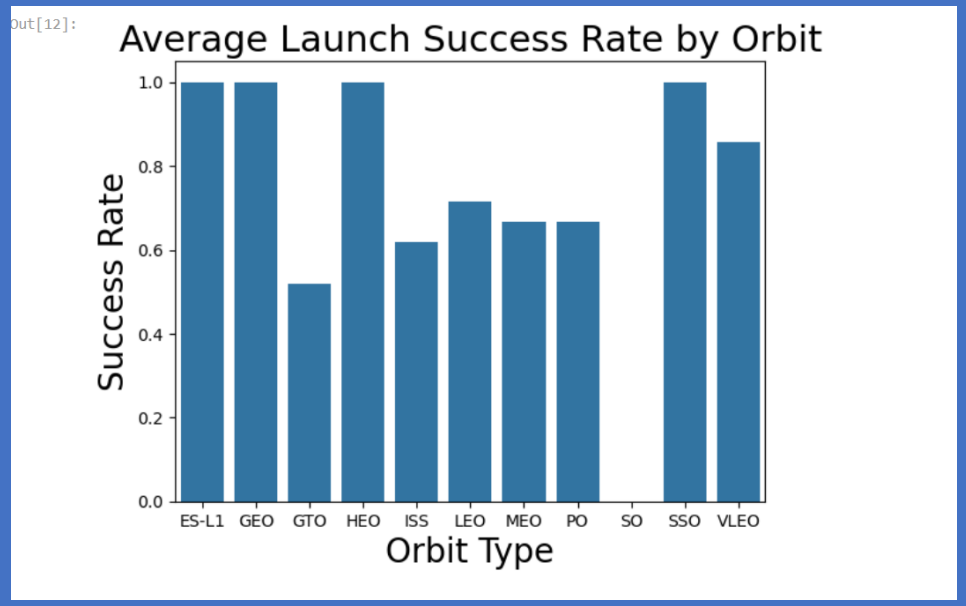
-Flight number and Launch Site
-Orbit type and Flight number
-Payload and Launch site
Payload and orbit Type



BAR CHART

Bar charts helped visualize the correlation between:

-Success rate and Orbit type



EDA with SQL

```
%%sql
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

Total_Payload
45596

```
%%sql
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

Average_Payload
2928.4

```
%%sql
SELECT MIN(Date) AS First_Success_GroundPad
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

First_Success_GroundPad
2015-12-22

EDA with SQL

```
%%sql
SELECT DISTINCT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

```
%%sql
SELECT Mission_Outcome, COUNT(*) AS Count
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

```
%%sql
SELECT DISTINCT Booster_Version, PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
);
```

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600

```
SELECT substr(Date, 6, 2) AS Month,
       Booster_Version,
       Launch_Site,
       Landing_Outcome
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE 'Failure (drone ship)%'
  AND substr(Date, 1, 4) = '2015';
```

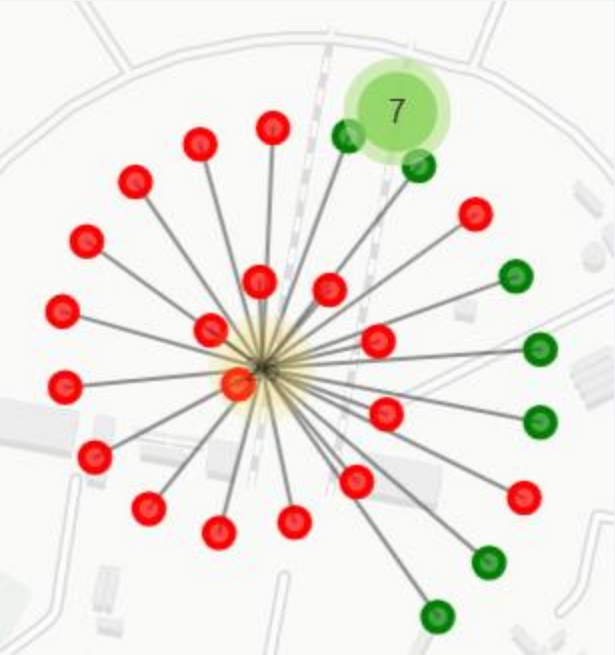
Month	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

```
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

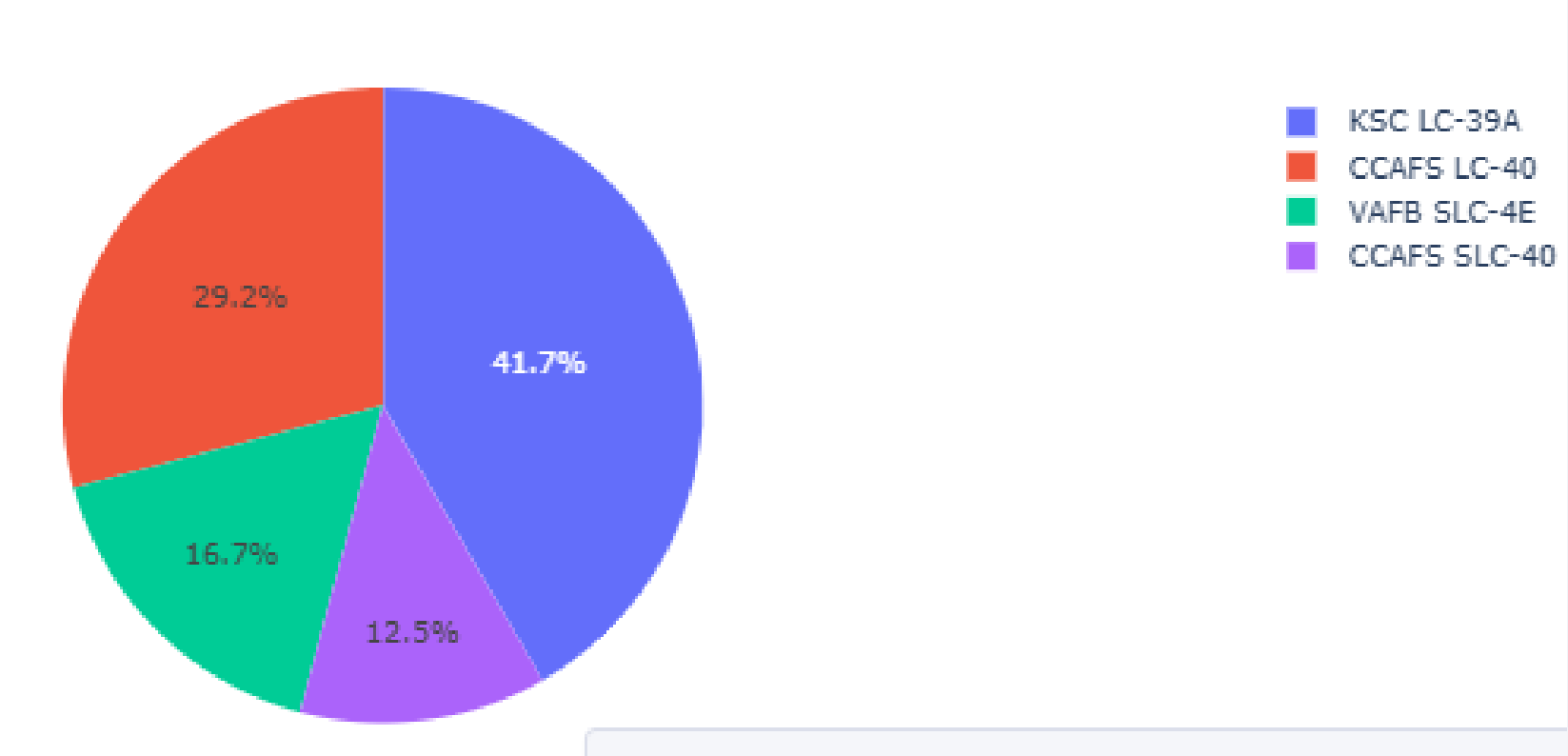
Build an Interactive Map with Folium

- First I created a Folium map object and used add folium.Circle and folium.Marker to represent the launch sites.
- Subsequently I created marker clusters to represent launches from the same site and color-code them depending on whether it was successful.



Build a Dashboard with Plotly Dash

- The pie chart makes it easy to see which sites have the most successful launches. It further has information about the individual launch site's success rate.



Predictive Analysis (Classification)

For model development, I used the train_transform_split() function to create training and testing data.

Then I created a GridSearchCV object. Using its output, I was able to determine hyperparameters and accuracy.

```
from sklearn.model_selection import train_test_split

X_train, X_test, Y_train, Y_test = train_test_split(
    X_scaled,
    Y,
    test_size=0.2,
    random_state=2,
    shuffle=True
)
```

Results

Exploratory Data Analysis Results

Interactive Analytics Demo in Screenshots

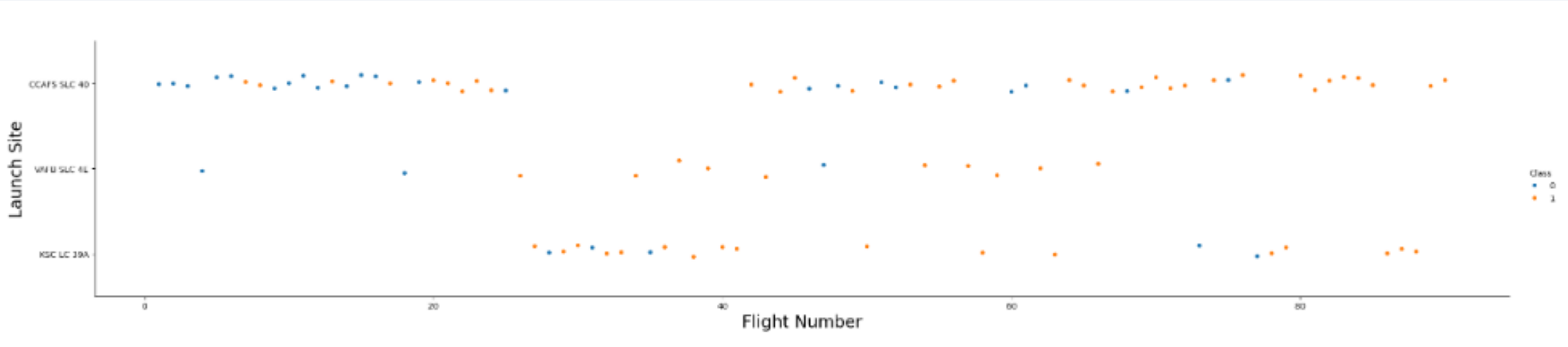
Predictive Analysis Results

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

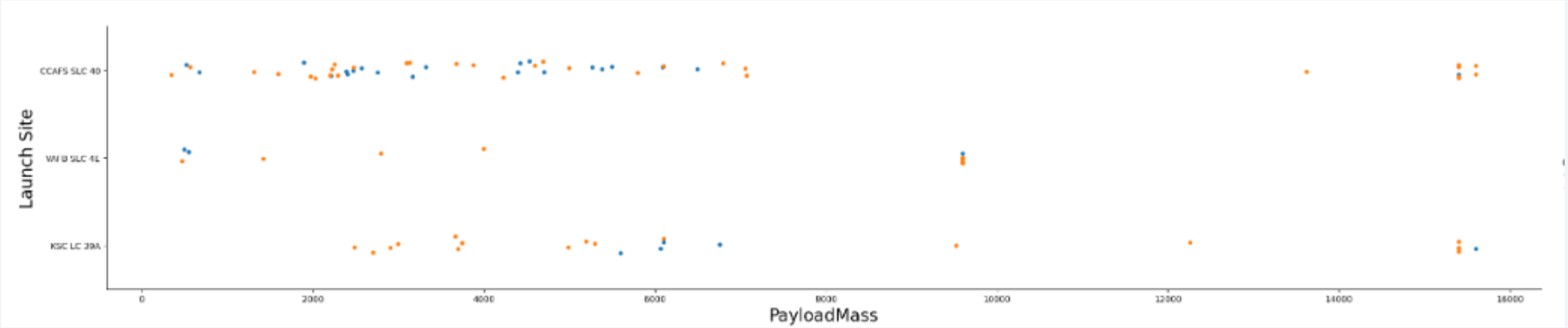
We can tell as the number of flight increases, so does the success rate. Most of the early flights launched at CCAFS SLC 40 were unsuccessful. There were no early flights at KSC LS 39A, so it appears as the launches from this site are more successful.



Flight Number vs. Launch Site

Payload vs. Launch Site

Payload vs. Launch Site



Success Rate vs. Orbit Type

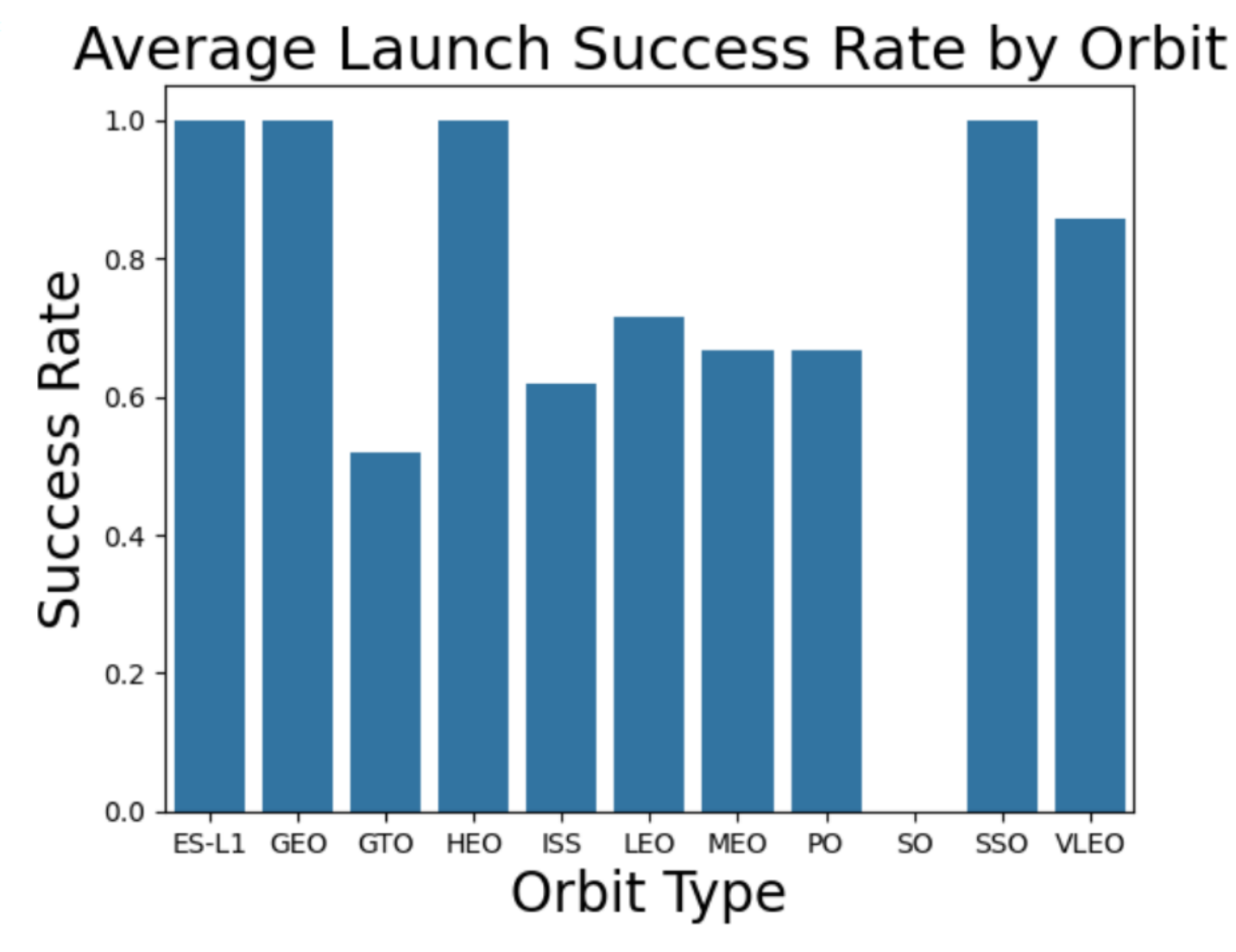
The orbits with the highest success rate are:

ES-L1

GEO

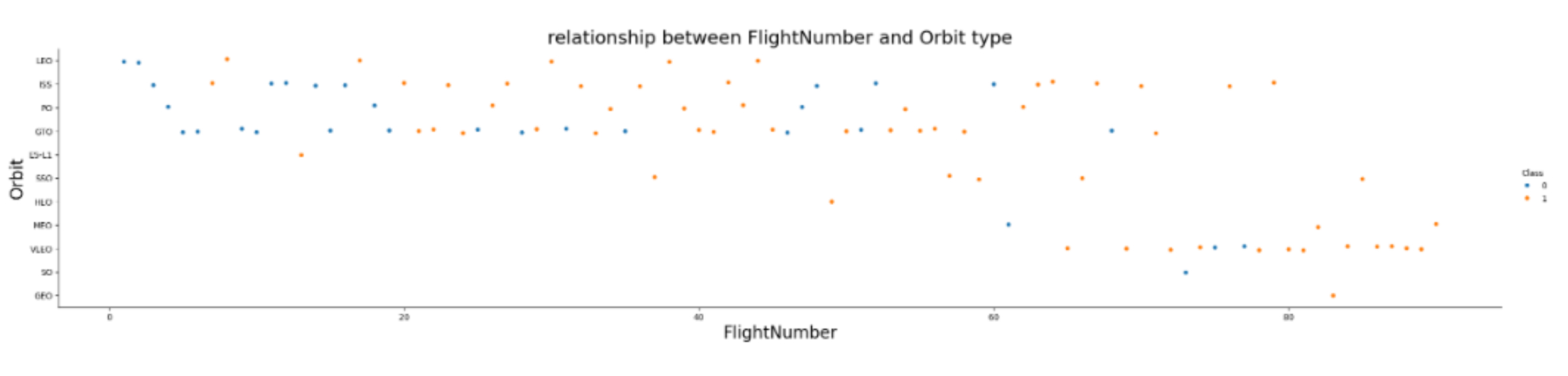
HEO

SSO



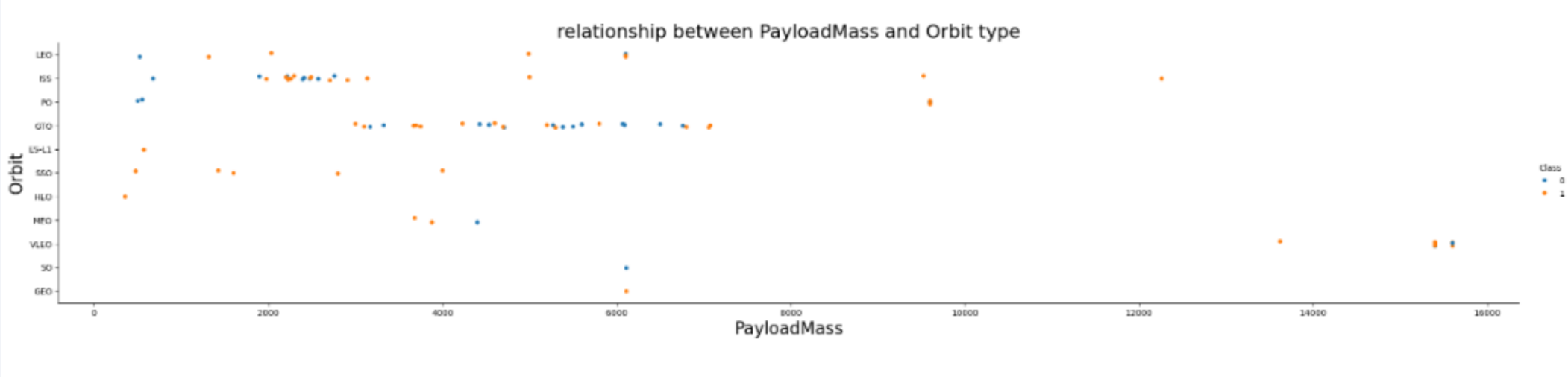
Flight Number vs. Orbit Type

We notice great success for VLEO, GEO and HEO orbits. However GEO and HEO have only one attempt. Therefore we cannot determine a correlation between orbits and flight number.



Payload vs. Orbit Type

ISS, LEO, po and VLEO experience success with a heavier payload.
However we can see in SSO, HEO and ES-L1 that the opposite is true.



Launch Success Yearly Trend

We see a success rate of 0 up until 2013, followed by a steady consistent climb hovering around 0.8 at present.

All Launch Site Names

- Names of the unique launch sites

```
%%sql
SELECT DISTINCT Launch_Site
FROM SPACEXTABLE;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- 5 records where launch sites begin with `CCA`

```
%%sql
SELECT *
FROM SPACEXTABLE
WHERE Launch_Site LIKE 'CCA%'
LIMIT 5;
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- Total payload carried by boosters from NASA

```
%%sql
SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload
FROM SPACEXTABLE
WHERE Customer = 'NASA (CRS)';
```

Total_Payload
45596

Average Payload Mass by F9 v1.1

- Average payload mass carried by booster version F9 v1.1

```
%%sql
SELECT AVG(PAYLOAD_MASS__KG_) AS Average_Payload
FROM SPACEXTABLE
WHERE Booster_Version = 'F9 v1.1';
```

Average_Payload
2928.4

First Successful Ground Landing Date

- Dates of the first successful landing outcome on ground pad

```
%%sql
SELECT MIN(Date) AS First_Success_GroundPad
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (ground pad)';
```

```
 * sqlite:///my_data1.db
Done.
```

First_Success_GroundPad
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- Names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%%sql
SELECT DISTINCT Booster_Version
FROM SPACEXTABLE
WHERE Landing_Outcome = 'Success (drone ship)'
  AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes

```
%%sql
SELECT Mission_Outcome, COUNT(*) AS Count
FROM SPACEXTABLE
GROUP BY Mission_Outcome;
```

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Names of the booster which have carried the maximum payload mass

```
%%sql
SELECT DISTINCT Booster_Version, PAYLOAD_MASS__KG_
FROM SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
    SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE
);
```

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

- Failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%%sql
SELECT substr(Date, 6, 2) AS Month,
       Booster_Version,
       Launch_Site,
       Landing_Outcome
FROM SPACEXTABLE
WHERE Landing_Outcome LIKE 'Failure (drone ship)%'
  AND substr(Date, 1, 4) = '2015';
```

Month	Booster_Version	Launch_Site	Landing_Outcome
01	F9 v1.1 B1012	CCAFS LC-40	Failure (drone ship)
04	F9 v1.1 B1015	CCAFS LC-40	Failure (drone ship)

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Ranked count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

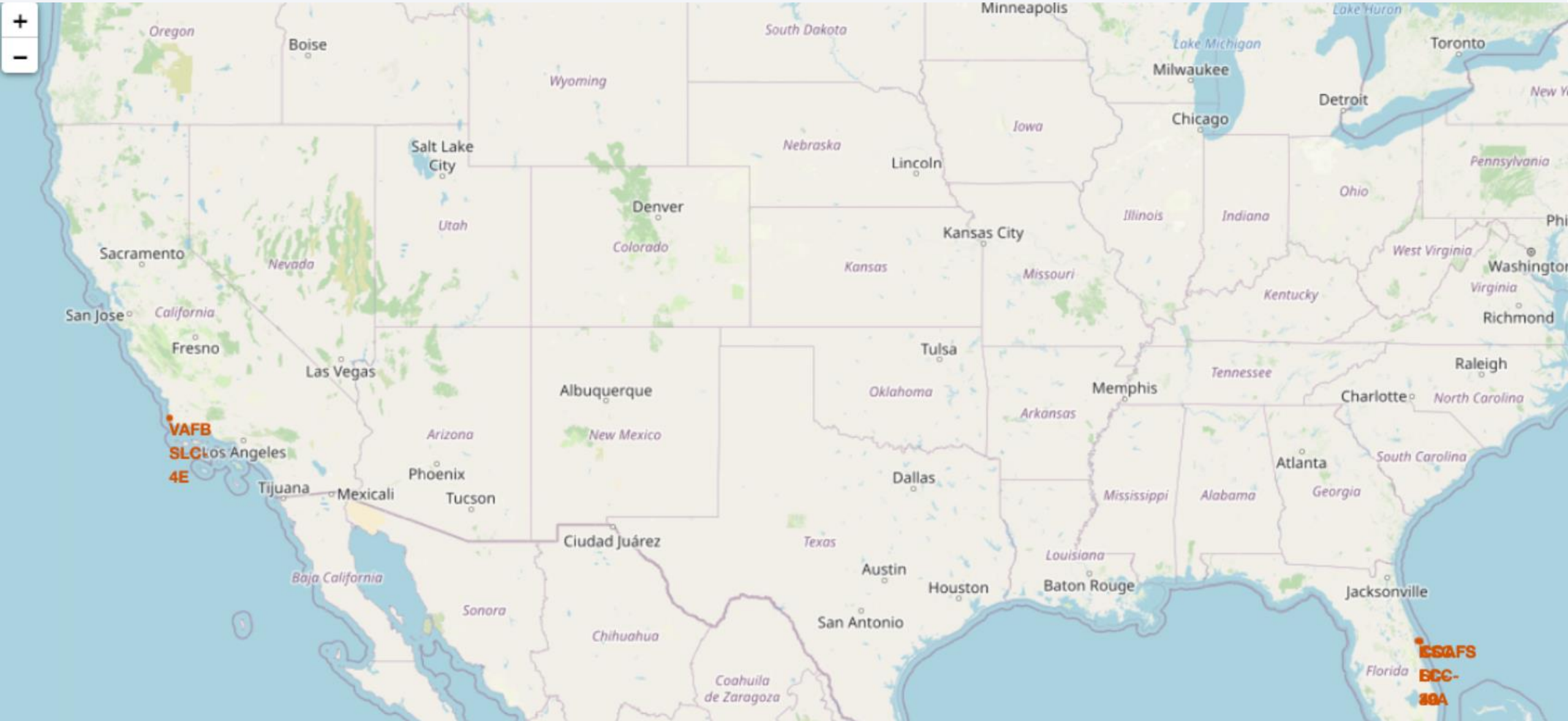
```
%%sql
SELECT Landing_Outcome, COUNT(*) AS Outcome_Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome
ORDER BY Outcome_Count DESC;
```

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

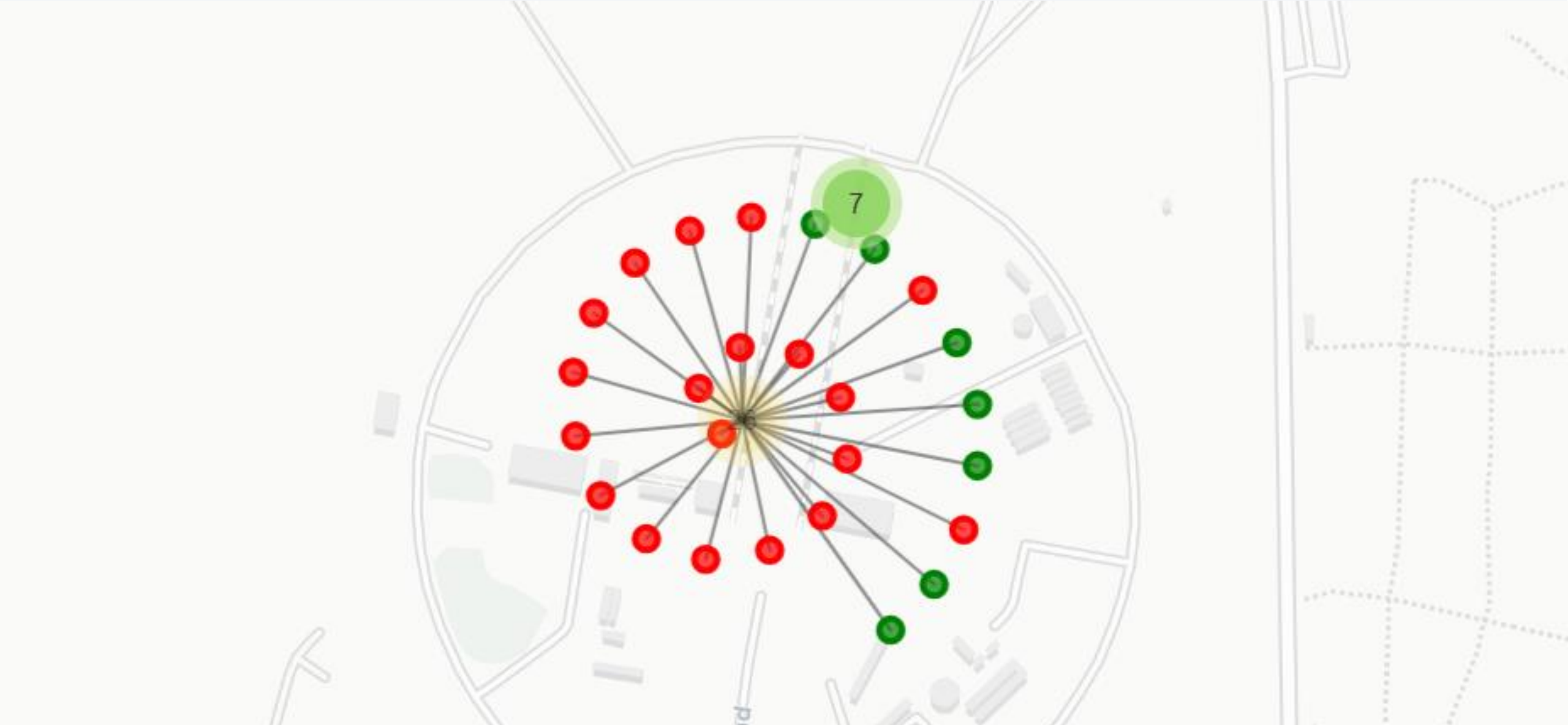
Section 3

Launch Sites Proximities Analysis

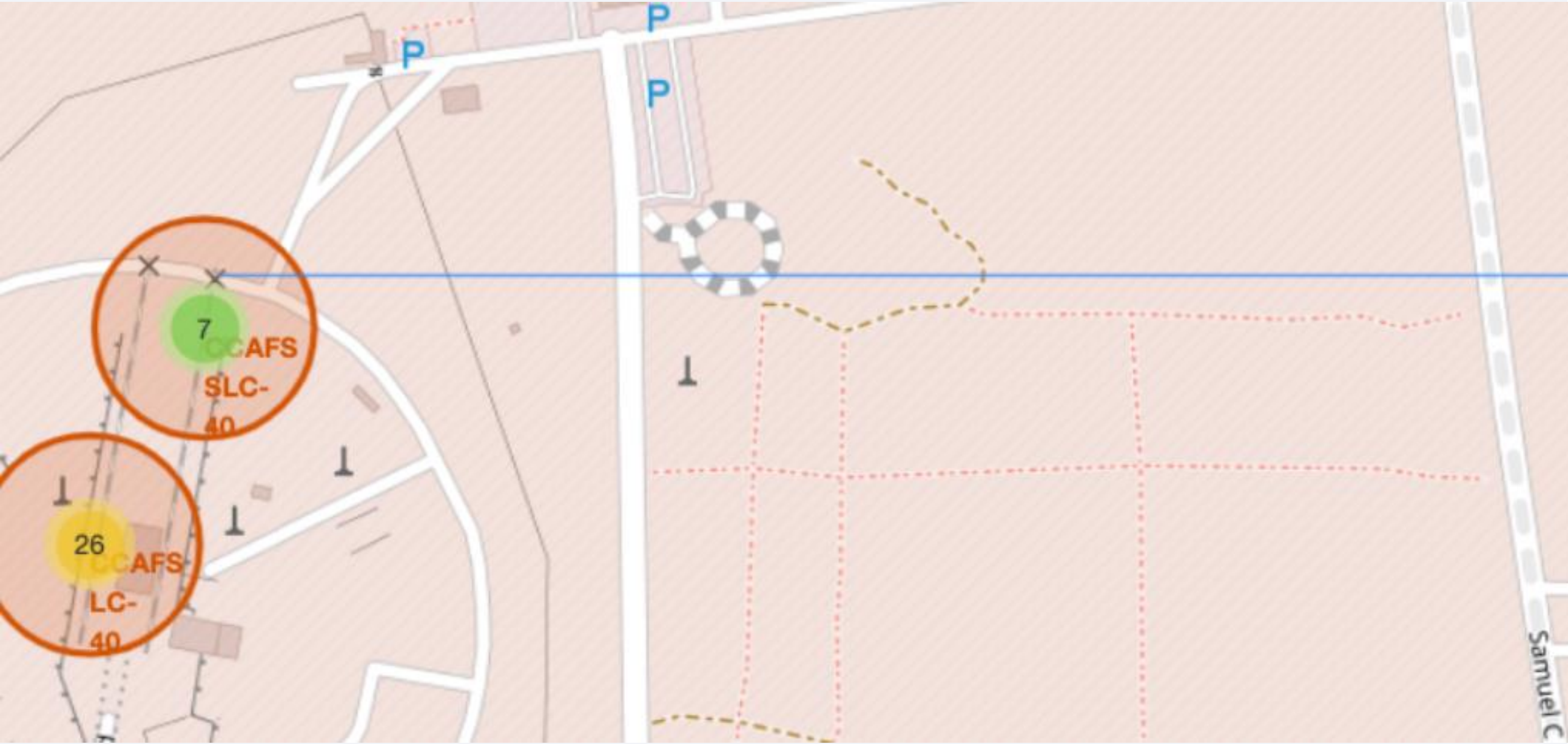
Folium Map Global View



Folium Map Color-labeled Launch Outcomes



Proximity of sites to infrastructure



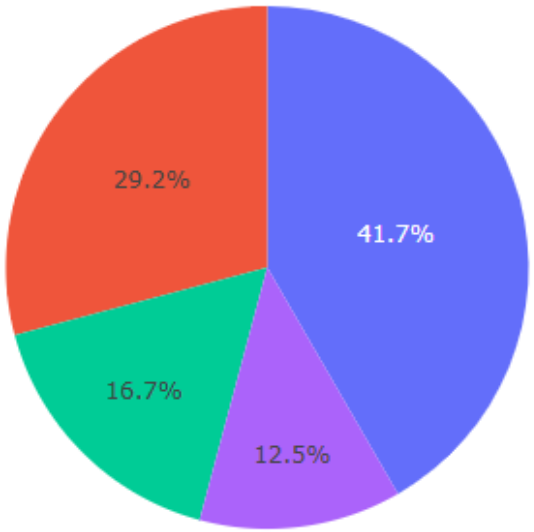
Section 4

Build a Dashboard with Plotly Dash

SpaceX Launch Records Dashboard

All Sites

Total Successful Launches for All Sites



KSC LC-39A
CCAFS LC-40
VAFB SLC-4E
CCAFS SLC-40

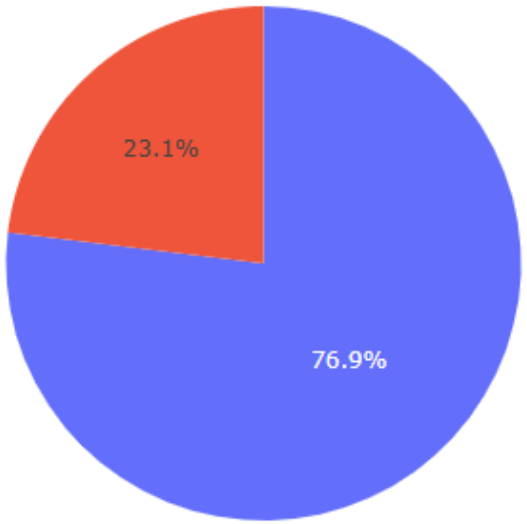
- KSC LC-39A has the highest success rate with 41.7%

SpaceX Launch Records Dashboard

KSC LC-39A

Success vs. Failure for site KSC LC-39A

- KSC LC-39 Launch site has the highest success vs failure ratio with a 76.9% for successful launches



1
0

Errors Callbacks v3.2.0 Server

Payload vs Success Scatter Plot

Payload range (Kg):

Section 5

Predictive Analysis (Classification)

Classification Accuracy

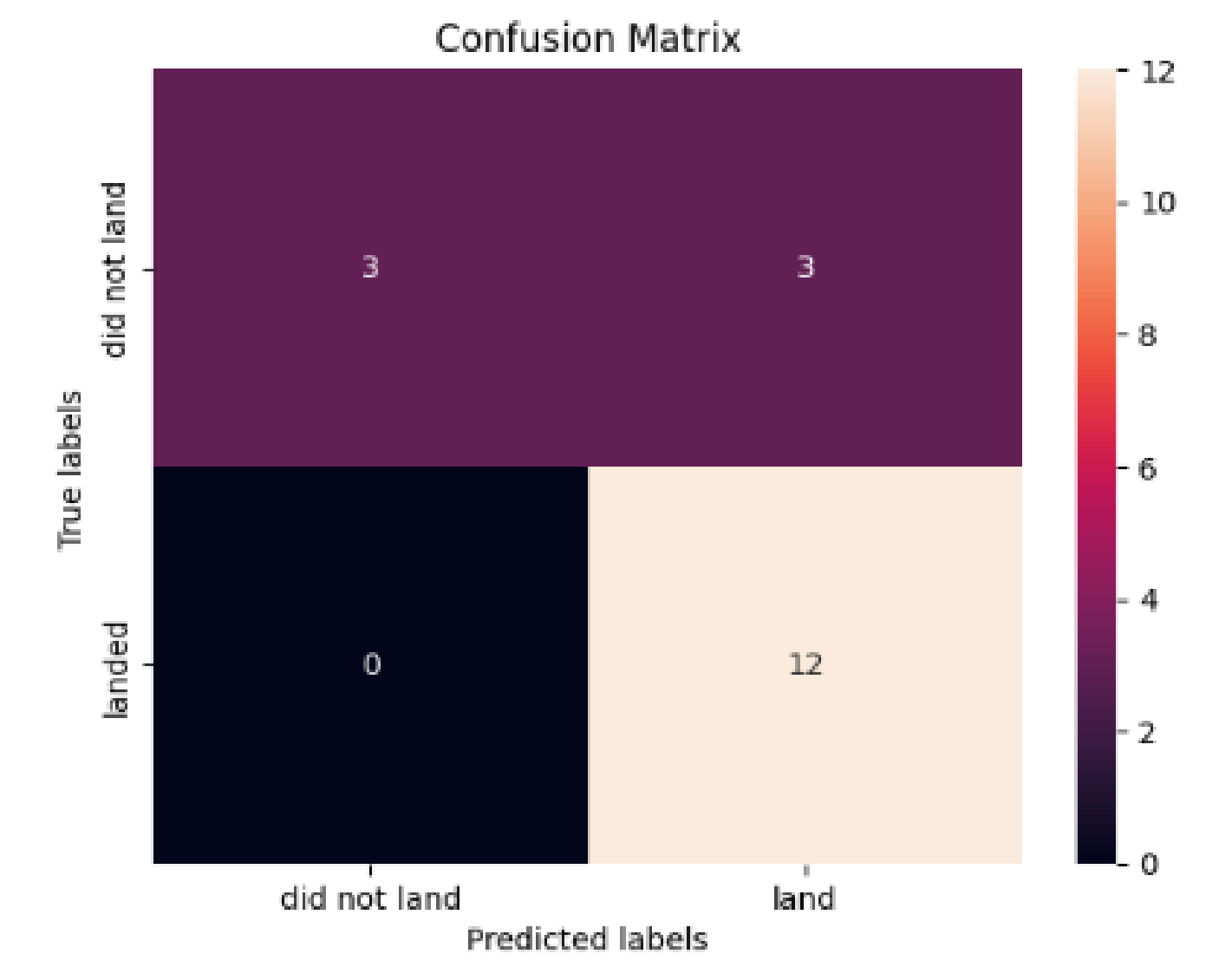
Plotting the Accuracy score and the Best score, we get the following results:

Find the method performs best:

```
[41]: print("Logistic Regression Test Accuracy:", logreg_cv.score(X_test, Y_test))
print("SVM Test Accuracy:", svm_cv.score(X_test, Y_test))
print("Decision Tree Test Accuracy:", tree_cv.score(X_test, Y_test))
print("KNN Test Accuracy:", knn_cv.score(X_test, Y_test))
```

```
Logistic Regression Test Accuracy: 0.8333333333333334
SVM Test Accuracy: 0.8333333333333334
Decision Tree Test Accuracy: 0.7777777777777778
KNN Test Accuracy: 0.8333333333333334
```

Confusion Matrix



Conclusions

- With the increase in flight numbers, the success rate also increases
- Orbits ES-L1, HEO, VLEO, GEO have the highest success rates. No correlation was observed when comparing payload to orbits.
- KSC LC-39A is the launch site with most successful attempts. However this may be explained with more flight numbers after having used and gathered data on other launch sites.

Appendix

Thank you!